

Supplementary material - Machine learning survival models for colorectal cancer: Advantages of Random Survival Forests over classification algorithms

Rafael Sant'Ana Herzog

2026-05-19

This document aims to present supplementary results that are referenced throughout the main text of the article “Machine learning survival models for colorectal cancer: Advantages of Random Survival Forests over classification algorithms”. Additionally, all scripts used and the resulting outputs can be found at [this GitHub repository](#).

1 Excluded covariates

As stated in the main text of the article, ten of the 25 covariates used by Buk Cardoso et al. (2023) were excluded from the analysis. Specifically, the following variables were removed:

- **IBGE** and **IBGEATEN**, which represented, respectively, the “IBGE code of the patient’s city of residence” and the “IBGE code of the institution”;
- **TRATHOSP**, which represented the “combination code of treatments carried out at the hospital”, but was a combination of other treatment variables already involved in the analysis;
- **NONE**, which represented “treatment received in hospital = none (0 = no and 1 = yes)”, but had very few observations in category 1 (only 0.3%);
- **TMO**, which represented “treatment received in hospital = tmo (0 = no and 1 = yes)”, but had only one observation in category 1;
- **IMUNO**, which represented “treatment received in hospital = immunotherapy (0 = no and 1 = yes)”, but had very few observations in category 1 (only 0.1%);
- **NENHUMANT**, which represented “treatment received outside hospital and before admission = none (0 = no and 1 = yes)”, but had very few observations in category 0 (less than 0.1%);
- **CONSDIAG**, which represented the “difference in days between the dates of consultation and diagnosis”, but had negative values;
- and **RRAS** and **DRS**, which represented, respectively, the “Regional Health Care Network code” and the “Regional Health Department code”.

2 Characteristics of the population in the total, training, and test datasets

The proportion of events was approximately the same across all total, train (70%) and test (30%) datasets, as shown in **Table 1**.

Event status	Proportion		
	Total dataset	Train	Test
0	0.588	0.59	0.586
1	0.412	0.41	0.414

Table 1: Proportion of events (failures) in the total, train, and test datasets.

The nonparametric estimated median survival time and the survival probabilities at 1, 3, and 5 years, presented in **Table 2**, were also roughly the same across the three datasets.

Time	Estimate		
	Total dataset	Train	Test
Median	6.55 (95% CI: 6.22–7.04)	6.7 (95% CI: 6.25–7.24)	6.31 (95% CI: 5.8–7.2)
1 year	0.815 (95% CI: 0.81–0.819)	0.814 (95% CI: 0.809–0.82)	0.816 (95% CI: 0.808–0.824)
3 years	0.622 (95% CI: 0.616–0.628)	0.621 (95% CI: 0.614–0.628)	0.625 (95% CI: 0.614–0.635)
5 years	0.535 (95% CI: 0.529–0.542)	0.537 (95% CI: 0.53–0.545)	0.532 (95% CI: 0.52–0.544)

Table 2: Estimated median survival and survival probabilities at 1, 3, and 5 years for the total, train, and test datasets.

Frequency distributions and log-rank test results for each covariate and dataset are summarized in **Table 3**. As shown, the percentages of each variable’s categories are roughly the same across the three datasets. The only differences in significance in the log-rank test were observed for the radiotherapy and hormone therapy variables.

Variable	Categories	%			P-value (Log-rank)		
		Total	Train	Test	Total	Train	Test
Education level	Illiterate	4.1%	3.9%	4.5%	<0.0001	<0.0001	<0.0001
	Incomplete	35.4%	35.4%	35.3%			
	Primary Education						
	Complete	36.6%	36.5%	36.7%			
	Primary Education						
	High School	17.4%	17.5%	17.2%			
Age group	Higher Education	6.6%	6.7%	6.4%	<0.0001	<0.0001	<0.0001
	0 to 49 years	17.2%	17.3%	16.8%			
	50 to 74 years	65.6%	65.5%	65.9%			
Sex	≥ 75 years	17.2%	17.2%	17.3%	<0.0001	<0.0001	0.0180
	Male	51.8%	51.8%	51.6%			
Diagnostic care category	Female	48.2%	48.2%	48.4%	<0.0001	<0.0001	<0.0001
	Agreement or Private Health-care	8.2%	8.2%	8%			
	Public Health-care	57.1%	57%	57.4%			
Previous diagnosis and treatment	No information	34.7%	34.7%	34.6%	<0.0001	<0.0001	0.0002
	No diagnosis / No treatment	41.9%	42%	41.7%			
Clinical staging group	With diagnosis / No treatment	58.1%	58%	58.3%	<0.0001	<0.0001	<0.0001
	I	12%	12.2%	11.8%			
	II	28.4%	28.4%	28.4%			
	III	32.9%	32.5%	34%			
Treatment received in hospital: surgery	IV	26.6%	27%	25.8%	<0.0001	<0.0001	<0.0001
	No	28.5%	28.7%	28.2%			
Treatment received in hospital: radiotherapy	Yes	71.5%	71.3%	71.8%	0.0415	0.3781	0.0179
	No	72.1%	72.1%	72%			
Treatment received in hospital: chemotherapy	Yes	27.9%	27.9%	28%	<0.0001	0.0004	0.0380
	No	29.5%	29.5%	29.3%			
Treatment received in hospital: hormone therapy	Yes	70.5%	70.5%	70.7%	0.0010	0.0055	0.0725
	No	99.3%	99.4%	99.3%			
Treatment received in hospital: others	Yes	0.7%	0.6%	0.7%	<0.0001	<0.0001	<0.0001
	No	92.6%	92.5%	93%			
Time between consultation and treatment	Yes	7.4%	7.5%	7%	<0.0001	<0.0001	0.0019
	≤ 60 days	75.1%	74.8%	75.8%			
Time between diagnosis and treatment	>60 days	24.9%	25.2%	24.2%	<0.0001	<0.0001	0.0039
	≤ 78 days	71.9%	71.6%	72.6%			
Year of diagnosis	>78 days	28.1%	28.4%	27.4%	<0.0001	<0.0001	<0.0001
	≤ 2006	21.2%	21.2%	21%			
Recurrence	>2006	78.8%	78.8%	79%	<0.0001	<0.0001	<0.0001
	No	87.5%	87.8%	87%			
	Yes	12.5%	12.2%	13%			

Table 3: Percentages and log-rank test results for each variable in total, train, and test datasets.

3 Additional context on the simulation study

To incorporate random censoring in the simulation study performed, censoring times were drawn from a uniform distribution over the interval $(0, \theta_h)$, for $h = 0, 1$. The value of θ_h was determined to achieve a desired right-censoring proportion p_c , by solving $P(T_h > C_h) = p_c$, where T_h and C_h denote failure and censoring times, respectively.

The survival behaviors of clinical stages I and II were specifically used as a basis because they ensured that all resulting θ_h values exceeded 5 — the longest time of interest in this study — across all evaluated scenarios. This guaranteed that censoring could occur throughout the entire follow-up period, allowing for a realistic distribution of censored observations.

The θ_h values obtained for each censoring probability and predictor are presented in **Table 4**.

Table 4: Values of theta for each censoring probability and predictor.

Theta / Censoring probability	0.25	0.40	0.50	0.60	0.70
Related to predictor $x = 0$	908.873	354.887	192.396	99.668	46.437
Related to predictor $x = 1$	242.288	97.841	54.306	28.878	13.880

References

Buk Cardoso, Lucas, Vanderlei Cunha Parro, Stela Verzinhasse Peres, Maria Paula Curado, Gisele Aparecida Fernandes, Victor Wünsch Filho, and Tatiana Natasha Toporcov. 2023. “Machine Learning for Predicting Survival of Colorectal Cancer Patients.” *Scientific Reports* 13 (1): 8874.